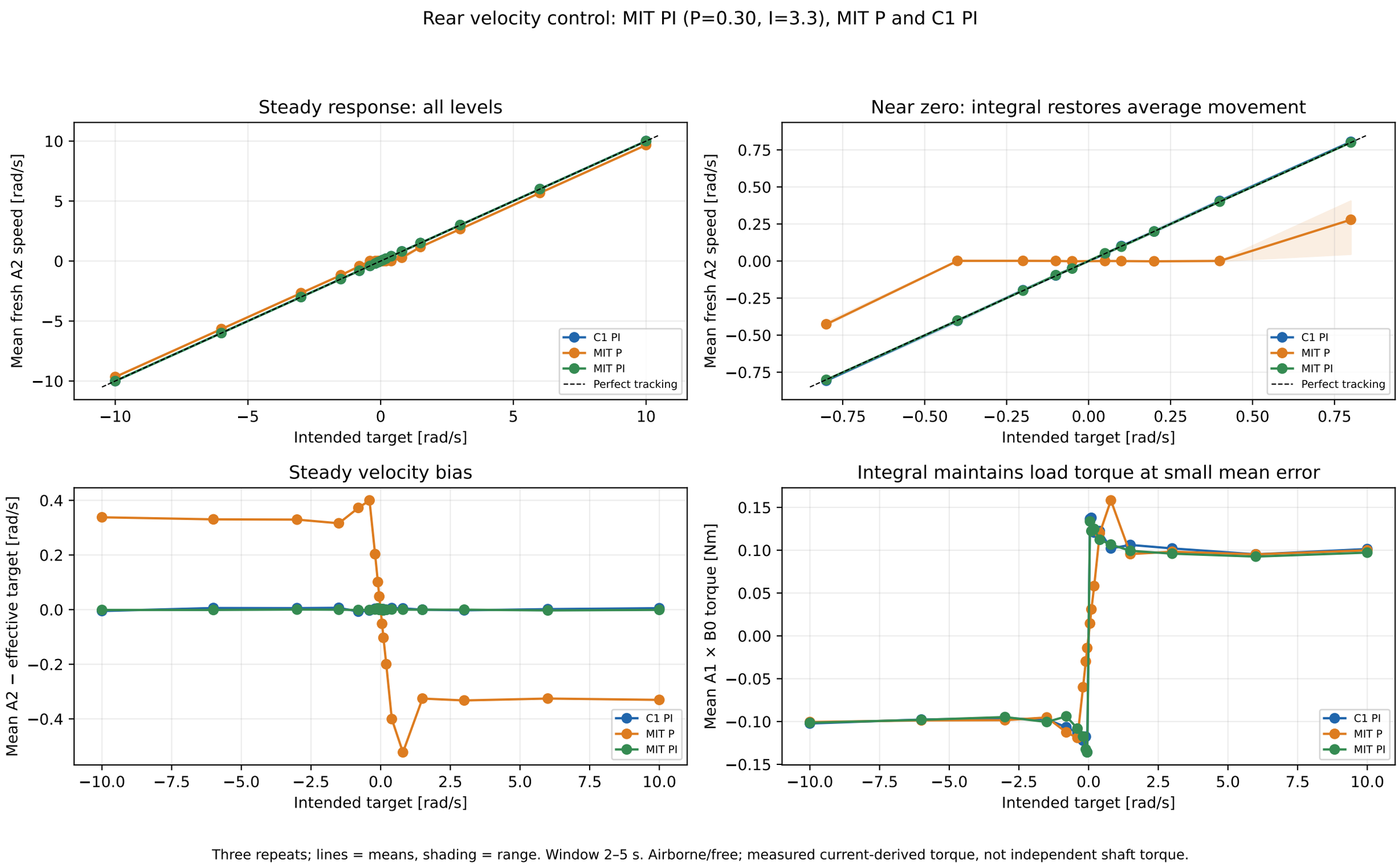
Rear velocity control: MIT PI (P=0.30, I=3.3), MIT P and C1 PI
Steady response: all levels
Mean fresh A2 speed [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Perfect tracking
Near zero: integral restores average movement
Mean fresh A2 speed [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Perfect tracking
Steady velocity bias
Mean A2 − effective target [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Integral maintains load torque at small mean error
Mean A1 × B0 torque [Nm]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Three repeats; lines = means, shading = range. Window 2–5 s. Airborne/free; measured current-derived torque, not independent shaft torque.